CS 5330 HW2

Questions

1. Give a couple of subjective explanations of what a filter can do.

   At the most basic a filter modifies pixels by sliding a kernel/window over the image. Sometimes it can take the average of surrounding pixels (blur), detect change in one direction (Sobel, Laplacian), or invert individual color values.

2. Give an example of a separable filter and explain the primary advantages of using them.

   A separable filter is a filter than can be split into smaller filters, but still gives the same result when they are convoluted. For a 2D 5×5 filter, this means that a 1×5 and a 5×1 filter can be convoluted and will result in the same 5×5 filter. The advantage to doing this is that the separable filter method is more computationally efficient, specifically O(n) instead of O(n^2).

3. What does it mean if a filter is an edge-preserving filter? Give an example of one.

   An edge preserving filter does some filter effect (i.e. blur) but also does so while preserving edges (i.e. edges will not be blurred). A common example is the median filter.

4. What is an example of a high-pass filter, and why would we want to use one?

   A blur filter is a high-pass filter. Its nature of averaging nearly pixels removes high-frequency noise in nearby pixel values and smooths out the entire image. Blur filters are useful to apply before other filters where high-frequency noise is troublesome, for example, edge detection filters. Since an edge detection filter essentially takes the gradient of the image, any noise is propagated. Removing this noise makes the resulting gradient estimation also have far less noise, giving a better edge detection.

5. What are a couple of ways of handling boundary issues (the edges of the image) when applying a filter?

   The three main solutions are 1) zero padding (extending the border of the image with black (zero) pixels so that the window slider can fit over all edge pixels. 2) Resizing the kernel on a case-by-case basis so that it fits over all edge pixels. And 3) Only applying the kernel over valid pixels and shrinking the output image.

6. You take the Fourier transform of an image and organize it so the DC and low frequencies are in the center of the visualization, and the high frequencies are around the outside of the image. If you were to set the values in a small circle at the center of the Fourier transform to zero and then convert the image back to the regular spatial domain, what would be the effect on the image?

   The result would be that the DC/low frequencies in the image would be removed. The resulting image would appear to have more high frequency "salt and pepper" noise and would be made sharper.

7. You have an image that consists of vertical and horizontal lines. After taking the Fourier transform of the image you set a centered horizontal bar of the Fourier image to zero, all except the central DC signal. After taking the inverse Fourier Transform, how do you think the resulting image would look?

   I expect that the vertical lines will disappear, but the horizontal lines will stay.